Assignment 1: Research, Plan, Design Document

GRIFFITH COLLEGE CORK

Student name:	**Karen Ferreira Magalhaes, Nátaly Fonseca Mendes, Percy Focazio-Moran, Rafiq Abudulai**		
Student number:	**3146094, 3123750, 3147979, 3104006**		
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Signed: Karen Ferreira Magalhaes, Nátaly Fonseca Mendes, Percy Focazio-Moran, Rafiq Abudulai
Date: **3 March 2026**

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1 Introduction

1.1 Description of the project and purpose of document (1-2 paragraphs)

1.1.1 This project proposes the development of an interactive international holiday calendar which allows users to explore public and cultural holidays from around the world. The web application will present holidays through an interactive and fun calendar interface, where users can browse by date, view detailed information about a specific holiday and search or filter holidays by country or type. Users will have the option to register and once registered, they will be able to personalise their experience by saving favourite holidays and even suggesting new holidays to be added to the calendar if they wish to. The application will mainly be developed using the MERN stack(MongoDB, Express.js, React.js and Node.js) and will support multiple users and sessions.

1.1.2 The purpose of this document is to outline our initial design and planning for the web application. It describes the main functionality of the system, design approach that will guide us through the development process and the target users. The document also presents interface design proposals such as wireframes and site maps. Together, these elements provide a rough and structured overview of the web application concept.

2 Functionality

2.1 Functional Requirements

2.1.1 Functional requirements are a key aspect of software development and system design. They define the specific behaviours and functions that a system or software should exhibit to meet the needs and expectations of its users.

2.1.2 Features:

1. User registration, login, logout. (Client: Authenticate forms, validation, protected routes. Server: create users, hash passwords).

2. Calendar browsing: month view + day details. (Client: interactive calendar grid, click a date to open the details panel. Server: endpoint returns holidays for a selected date, month and country).

3. Country, type filters + holiday search (Client: filter controls, live results. Server: query database using filters and text search; return matching holidays).

4. Holiday details page. (Client: details UI as name, date, description, tags. Server: fetch holiday by ID).

5. Favourites (Client: favorite holidays, favourites list, highlight favourites on calendar. Server: store favourites per user; endpoints to add, remove, list favourites).

6. Community suggestions. (Client-side: suggest form, upload reference link, status tracking: pending, approved. Back-end: save suggestions, admin endpoints to approve or reject, write approved changes into the holidays collection).

7. Admin dashboard. (Client-side: admin-only page, tables, filters, approve/reject buttons. Back-end: role-based access control, endpoints to list suggestions/users).

3 Research

3.1 User Analysis

3.1.1 The target users for this web application are individuals who need a simple and organised way to view holidays and manage important dates from different countries. The system is designed for a wide audience including students, workers, travellers, and people interested in international events and cultural celebrations.

The primary user group is expected to be between 18 and 40 years old, as this demographic frequently uses digital tools and web applications to organise schedules and plan activities. The application is designed to be gender neutral, meaning it does not target any specific gender and aims to be inclusive for all users.

Users are likely to access the site using both desktop and mobile devices, so the interface must be simple, responsive, and easy to navigate. Typical users may want to:

- Check holidays in different countries
- Search for specific holidays
- Save favourite holidays
- Suggest new holidays to the community
- Quickly view holidays in a calendar format

The system should therefore prioritise accessibility and simplicity, ensuring that users can find relevant information quickly without needing technical knowledge.

3.2 Personas

3.2.1 **Name:** Sarah

Age: 21

Occupation: University Student

Sarah is a university student who enjoys travelling and learning about different cultures. She plans trips during holiday periods and likes to keep track of international celebrations and public holidays. She likes to use calendar applications to organise her schedule and prefers simple and visually clear tools that are also aesthetically pleasing and allow customisation.

Goals:

- Easily check holidays in different countries
- Plan trips around international holidays
- Save interesting holidays to view later
- Quickly search for specific events

Frustrations:

- Websites that are cluttered or difficult to navigate
- Having to search multiple websites for holiday information
- Lack of filtering or search functionality

How the website helps her:
The calendar website allows Sarah to quickly browse holidays by date, search for specific holidays, and save favourites. The filtering system also allows her to view holidays from specific countries when planning trips.

3.2.2 **Name:** Daniel
Age: 32
Occupation: Remote Software Developer

Daniel works remotely with international teams. Because his colleagues are located in different countries, he often needs to check public holidays in various regions to coordinate meetings and project timelines.

Goals:

- Quickly view holidays in multiple countries

- Avoid scheduling meetings on public holidays
- Keep track of important international dates

Frustrations:

- Not knowing when colleagues are on national holidays
- Difficult-to-read calendar interfaces

How the website helps him:
The filtering system and calendar view allow Daniel to easily identify holidays in different countries, helping him better organise meetings and work schedules.

3.3 Competitive Analysis

3.3.1 Similar Projects

Prior to developing our calendar application we researched several existing websites that provide similar functionality to source some design inspiration. Our aim was to understand how other platforms display calendar information, how users interact with the interface, and what features are users expect in calendar applications.

The research focused primarily on websites that present event calendars, international holidays, or festival listing, as these are most similar to the proposed project. Additionally, several creative and portfolio websites were analysed for visual design and interactive elements that could improve the user experience of our project.

We identified common features such as:

- Calendar views organised by month or date
- Lists of events or holidays
- Search and filtering options
- Interactive UI elements
- Visually engaging layouts and colour schemes

Understanding how these features are implemented in existing websites helps guide the design decisions for our project and ensures the final product meets the expectations of the average user.

3.3.2 Reference Websites

During the research phase we analysed multiple websites to understand both the functional features of calendar systems and the visual design approaches used in modern web applications. These websites were grouped into two main categories: calendar and event-based websites and design inspiration websites.

Calendar and Event-Based Websites

The following websites were analysed because they provide functionality similar to the proposed project, such as displaying events or holidays organised by date. These websites demonstrate different approaches to presenting calendar information and managing event listings.

Websites researched:

- https://onlinecalendar.site/
- https://fashioncalendar.com/#calendar
- https://www.thatfestivalsite.com/festivals/calendar
- https://www.un.org/en/observances/list-days-weeks
- https://www.visitshropshire.co.uk/whats-on/

These websites typically provide features such as:

- Calendar views organised by month or date
- Lists of events or holidays associated with specific days
- Search and filtering tools to help users find events
- Detailed pages for individual events or holidays

Our analyses/research of the above websites taught us that successful calendar systems prioritise clear date navigation, readable layouts, and quick access to event details. These ideas influenced the design of our project, particularly the use of a monthly calendar view with clickable dates and detailed holiday information pages.

Design Inspiration Websites

In addition to functional calendar websites, several creative and interactive websites were researched to gather inspiration for the **visual style and user experience** of the project.

Websites researched:

- https://chromia.com/
- https://36.bienal.org.br/en/
- https://yes-dancebiennale.dk/en-gb
- https://www.mr-pandas-psychologically-safe-portfolio.com/
- https://mystella.me/ja
- https://chloeschumacher.com/
- https://garagemca.org/en
- https://www.wallofportfolios.in/
- https://www.awwwards.com/

These websites were examined for their creative design elements and interactive features, including:

- Animated or moving page elements
- Mouse-based interaction effects
- Unique layouts and typography
- Colour schemes and visual themes

From our analysis of the above websites, we gathered ideas to help us create a more visually engaging calendar interface whilst still maintaining usability and accessibility for users.

3.3.3 Comparison and Key Insights

From comparing the researched websites, several common design patterns and usability principles became clear. Most calendar and event websites focus on clear visual organisation of dates, simple navigation, and easy access to detailed information about each event or holiday. However, many of these platforms prioritise functionality over visual engagement.

In contrast, the design-focused websites emphasise interactivity, animation, and creative layouts, which make the browsing experience more engaging but sometimes reduce usability if overused.

The goal of our project is to combine the strengths of both categories by creating a calendar system that is functional and easy to navigate, while also incorporating visual elements and interactive features that improve the overall user experience without distracting from the main purpose of the site.

3.4 Content Inventory

3.4.1 A content inventory identifies and organises all the main types of content that will appear on the website which helps ensure that the site structure is clear and that all required information is planned before development begins. This website will contain several key sections which may include user authentication, the calendar interface, holiday information pages, and community interaction features.

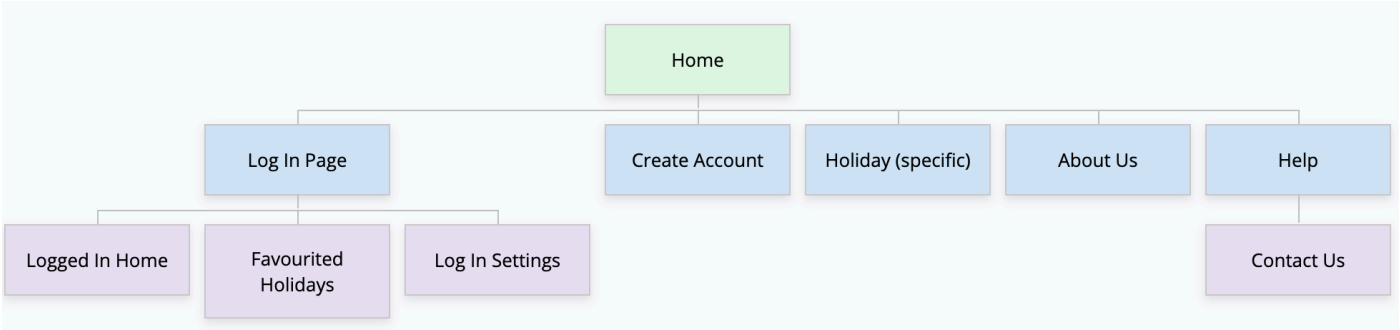
Page / Section	Content Description
Home Page	Main landing page containing the interactive calendar view, search bar, login button, and quick access to holiday information.
Calendar View	Monthly calendar displaying dates and highlighting holidays. Users can click on a specific date to view more details.
Holiday Details Page	Displays detailed information about a selected holiday, including name, date, country, description, and category tags.
Search and Filters	Allows users to search for holidays by name and filter results by country or holiday type.
User Registration	Form where new users can create an account by providing necessary information such as email and password.
Login / Logout	Authentication system allowing registered users to access personalised features and securely log out.
Favourites	A list of holidays saved by the user. Users can mark holidays as favourites directly from the calendar or holiday page.
Holiday Suggestion Form	Form that allows users to suggest new holidays or events to be added to the system.
Admin Dashboard	Administrative interface used to review user suggestions, approve or reject submissions, and manage users.

4 Design specifications

4.1 Site map

4.1.1 For the site map, we used the free online tool called Gloomaps.

Site map can also be found at https://www.gloomaps.com/rzsDrslk4o



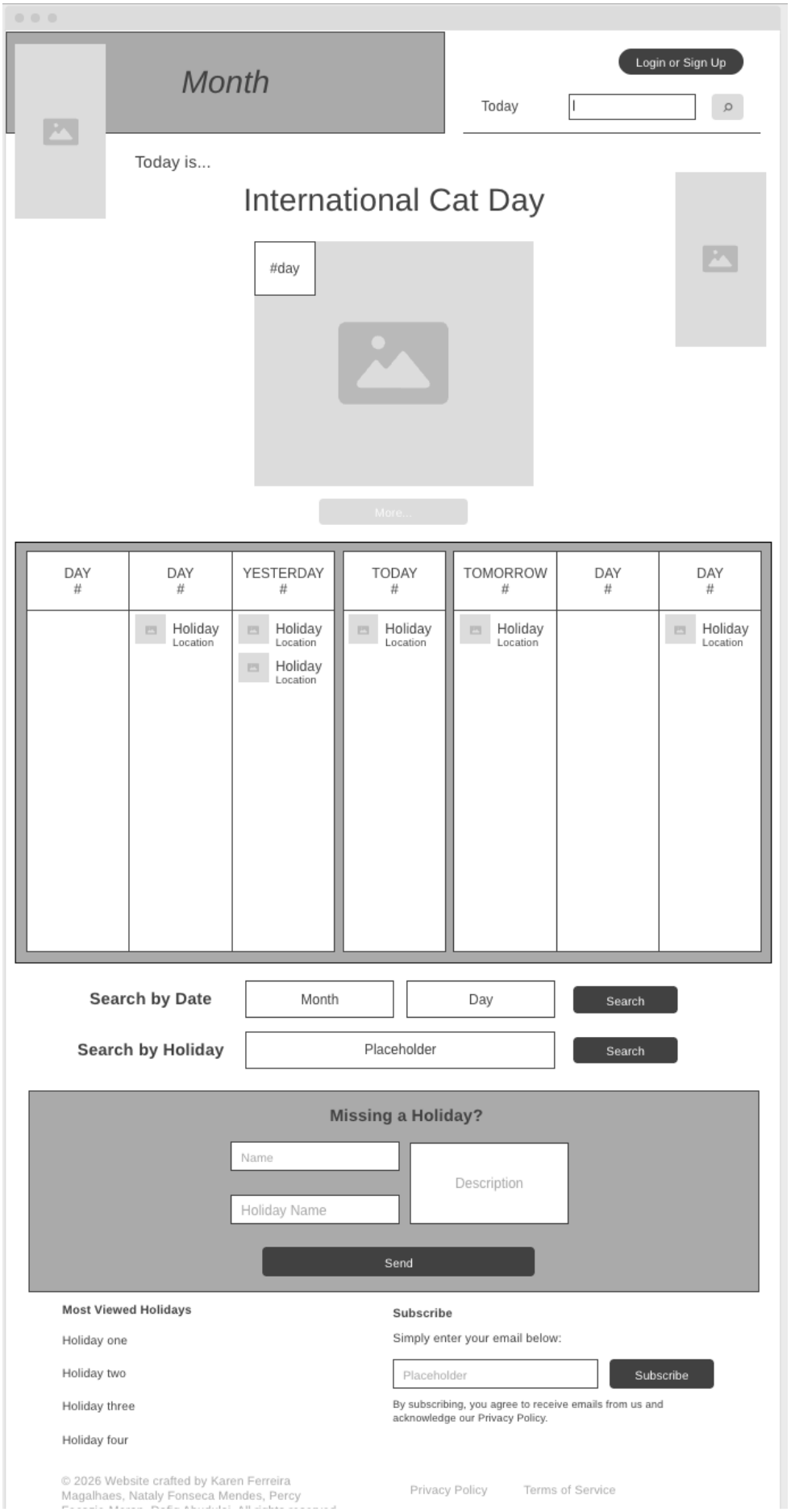
Site Map

4.2 Wireframe templates

4.2.1 The group utilised Wireframe.cc as the main wireframe creation device. The following page is a wireframe of the home page, which includes the following features:

- ➔ Search by Date
- ➔ Search by Holiday (two places)
- ➔ Log in button
- ➔ Holiday recommendation form
- ➔ Subscription box
- ➔ List of holidays on each day (utilising database)

Wireframe can also be found at https://wireframe.cc/xSBUnr

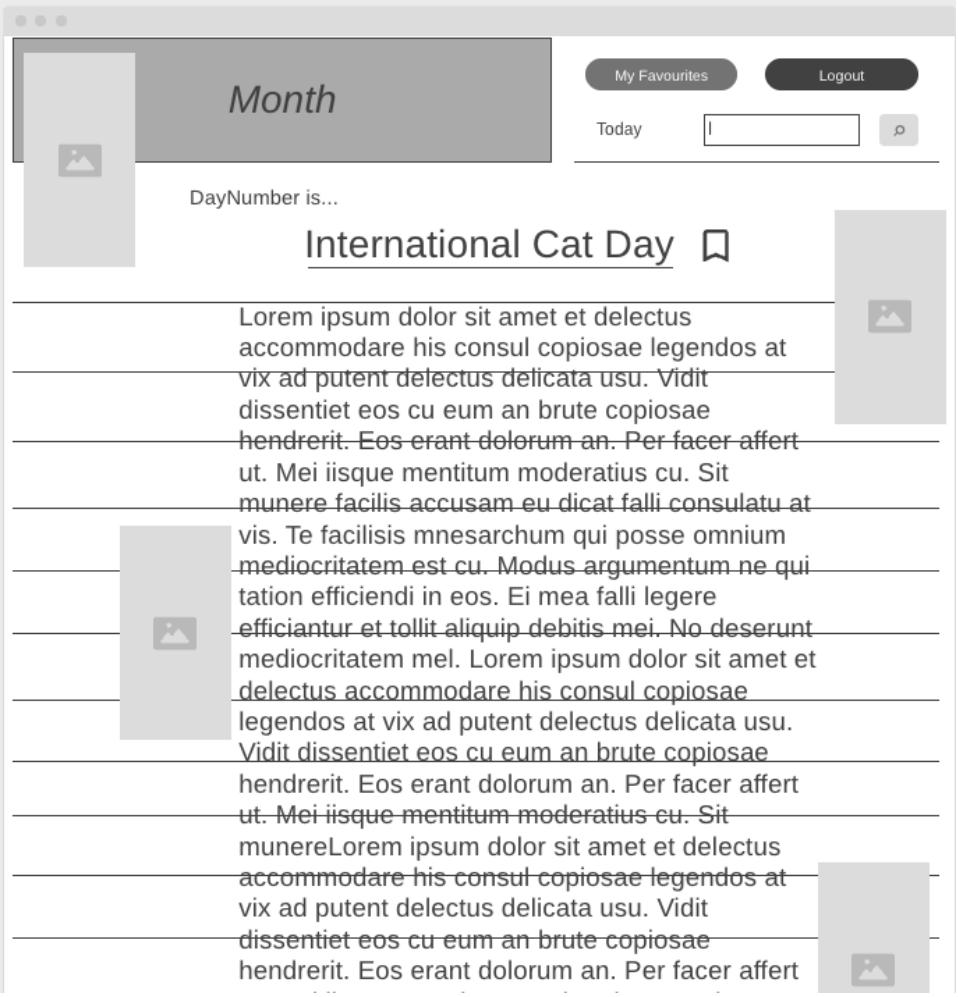


Wireframe of home page

4.2.2 The following image is a wireframe of a holiday description page with the user logged in, which includes the following features:

- ➔ Log out button
- ➔ Favourited holiday list
- ➔ Ability to favourite holidays
- ➔ Search button
- ➔ Link to main home page

Wireframe can also be found at https://wireframe.cc/iVndf6



Wireframe of holiday page with user logged in

4.3 Colour

Colour is vital for creating the stationary effect we want for the international calendar. Additionally, we need to ensure that our site is accessible and able to be viewed by those with reduced sight or colour-blindness. The palette is created to bring the feel of paper, ink, and highlighter. It features soft, soothing background tones, while a selection of bold “ink” colours is reserved for text and UI elements.

As a result, we have chosen the following colours for our site:

4.3.1 Our background colour will be:

- Floral white: #FAF7F0 (main page background);
- Cream: #FFFDF8 (cards, calendar sheets);
- Bone: #E7DED0 (borders, subtle dividers).

4.3.2 Our text colour will be:

- Jet Black: #1F2937 (Primary text, near-black, comfortable on warm paper)
- Charcoal Blue: #374151 (Secondary text, subheadings, metadata)
- Slate Grey: #6B7280 (helper text)

4.3.3 Our accent colour will be:

4.3.3.1 Buttons:

- Royal Azure: #1D4ED8 (buttons, links)
- Persian Blue: #1E40AF (buttons, links hover)

4.3.3.2 “Highlighter” tones (use for tags, and small details):

- Jasmine: #FDE68A
- Aquamarine: #A5F3DF
- Periwincle: #C6BCF4

- Light Caramel: #FDBA74

4.3.3.3 Status:

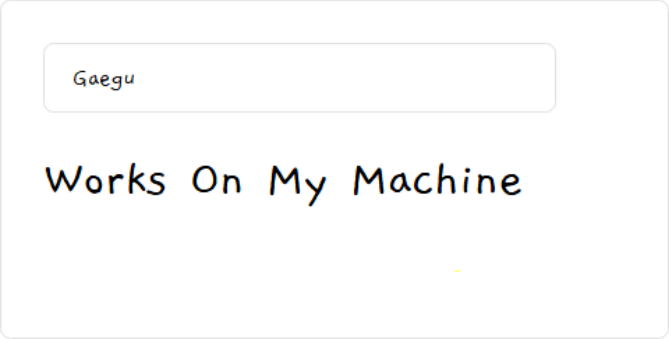
- Turf Green: #1A753D (success)
- Autumn Leaf: #EA6D1F (warning)
- Primary Scarlet: #DF2A2A (error)
- Baltic Blue: #075985 (information)

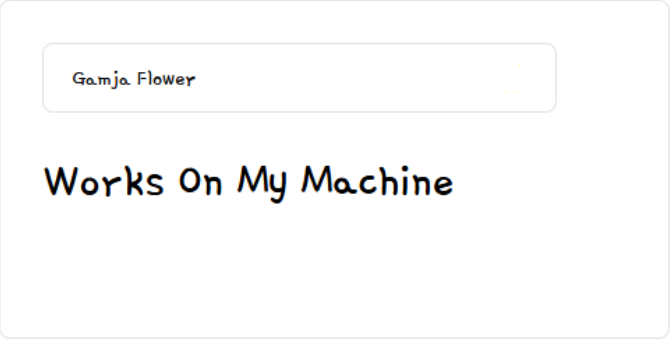
4.4 Fonts

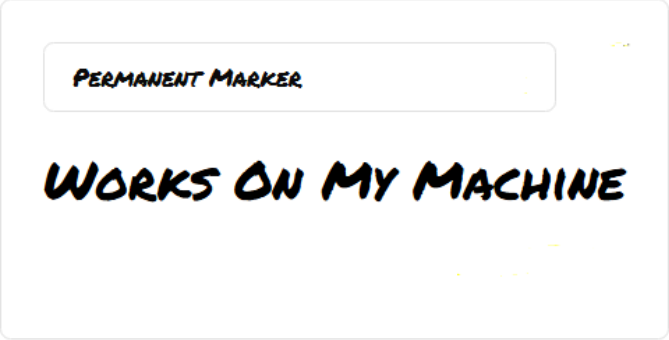
4.4.1 The site is designed to emulate stationary, and the font is designed to do the same. Some handwritten fonts are difficult to read, however, so we were considerate in our decision to pick multiple fonts through the site based on legibility and amount of text. For example, the Month can be in a cursive script, as it is one word, but for the informational text of holidays, it needed to be more able to be read at a glance by anyone. Similarly, the log in and button fonts needed to be easy to read.

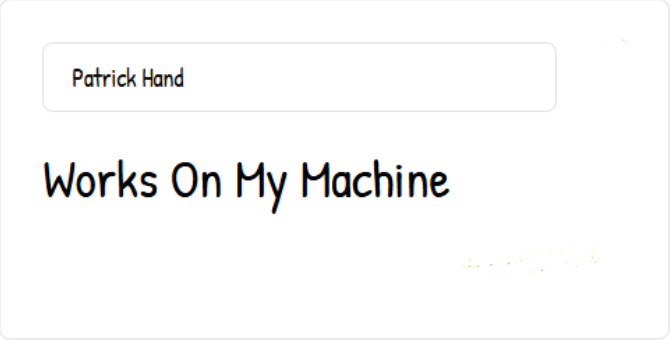
4.4.2 As a result, the fonts chosen are the following:

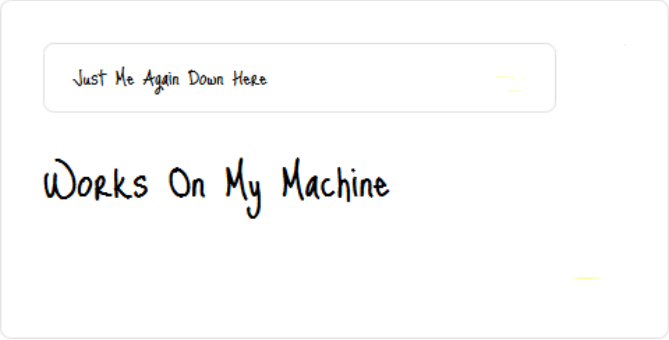
➔ Header: Options of font as Gaegu, Gamja Flower, Permanent Marker, Patrick Hand, Just Me Again Down Here. All of them are free and available in Google Fonts, Compared using Font Comparison Tool.



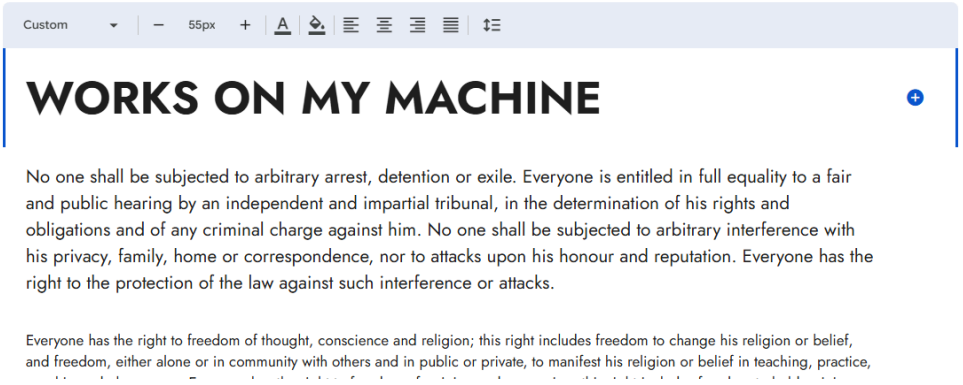






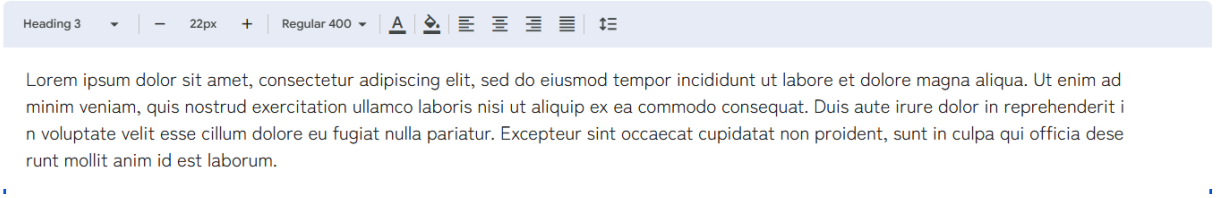


➔ Title: Based on one of our web application references, the options for text are:

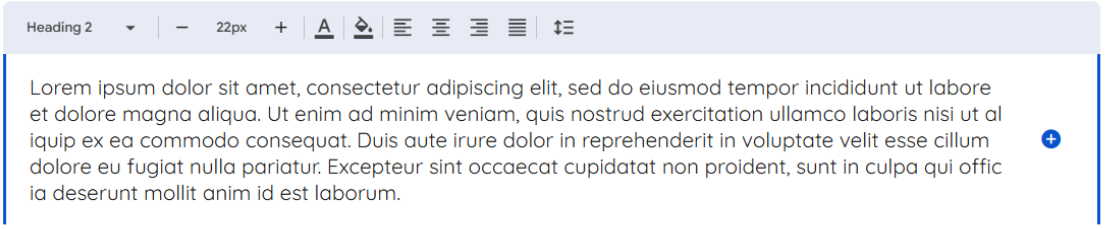


Jost: "Is an original font created by indestructible type*. It is inspired by 1920s German sans-serifs. This is version 3.7. Jost is designed and maintained by Owen Earl, who is the creator of the font foundry indestructible type*. in 2020 Owen Earl, and Mirko Velimirovic worked together to make Jost a variable font."

➔ Text: Based on one of our web application references, the options for text are:



Zen Maru Gothic: "Is a rounded san serif family that gives a soft and natural impression due to the deep rounds in the corners. Because of this unique soft impression the thin weight is also easy to use in any scenes. Cute and fashionable, soft and easy for any communication."



Quicksand: "Is a display sans serif with rounded terminals. The project was initiated by Andrew Paglinawan in 2008 using geometric shapes as a core foundation. It is designed for display purposes but kept legible enough to use in small sizes as well. In 2016, in collaboration with Andrew, it was thoroughly revised by Thomas Jockin to improve the quality. In 2019, Mirko Velimirovic converted the family into a variable font."

5 The project timeline/Schedule

Below we have provided a mock up of our task division plan over time. This is just to give an idea of the timeline and anticipated tasks. Further division of tasks in a separate document, as this is extremely rough division, and we will likely help each other through all tasks. The Project_Steps.pdf was used to draft the steps.

Week	**Task**	**Assigned**
3 - ending in 22/02/26	Brainstorm ideas for project Create Repository	Karen, Percy Karen
4 - ending in 01/03/26	Research of references, initiated documents Wireframe initial concepts, organised assignment 1 draft Group discussion	Karen Percy Karen, Nátaly, Percy, Rafiq
5 - ending in 08/03/26	Features suggestion Organise references, screenshot and explain to new group members concept Final wireframes and sitemap for assignment 1 Research section Initiation form lead (help from all group members) Colour and Font research **Assignment 1: Design Document (due 08/03/26)**	Karen, Nátaly Percy Percy Rafiq Nátaly Karen Karen, Nátaly, Percy, Rafiq
6 - ending in 15/03/26	Segregating data decisions, drafting tables needed, securing database Provide HTTP endpoints for your frontend, which allow it to operate on your data. E.g. Create, Read, Update and Delete (“CRUD”) records. Authenticate users (verify they are who they say they are: aka log them in). Authorization. When a logged in user makes a request, the backend will determine whether they are allowed (authorized) to perform the requested action. Serve the frontend	Karen, Nátaly, Percy, Rafiq
7 - ending in 22/03/26	Unsure of what needed components will be **Assignment 2: Server-side components (due 23/03/26)**	 Karen, Nátaly, Percy, Rafiq
8 - ending in 29/03/26	Build HTML pages	Karen, Nátaly, Percy, Rafiq
9 - ending in 04/04/26	Use React (modern JS) to animate pages	Karen, Nátaly, Percy, Rafiq
10 -	Unsure of what needed components will be	

ending in 12/04/26	**Assignment 3: Client-side components (due 13/04/26)**	Karen, Nátaly, Percy, Rafiq
11 - ending in 19/04/26	Host, deploy site Polish, fix any bugs	Karen, Nátaly, Percy, Rafiq
12 - ending in 26/04/26	Unsure of what needed components will be **Assignment 4: Finalise project deliverable & Presentation of project (due 27/04/26)**	Karen, Nátaly, Percy, Rafiq

References

Create a palette - coolors (04 March 2026).

https://coolors.co/faf7f0-ffffd8-e7ded0-1f2937-374151-6b7280-1d4ed8-1e40af-075985.

Create a palette - coolors (04 March 2026).

https://coolors.co/1a753d-ea6d1f-df2a2a-a5f3cf-c6bcf4-fdba74-fde68a.

Font comparison Tool (06 March 2026). https://fontscompare.com/?share=eyJzYW1wbGVUZXh0IjoiV29ya3MgT24gTXkgTWFjaGluZSIsImZvbnRTaXplIjozMCwibGluZUhlaWdodCI6MS41LCJmb250U3R5bGUiOiJub3JtYWwiLCJ0ZXh0Q29sb3IiOiIjMDAwMDAwIiwiZm9udEJveElkcyI6WyIxIiwiMiIsIjMiLCI0IiwiNSJdLCJzZWxlY3RlZEZvbnRzIjp7IjEiOiJHYWVndSIsIjIiOiJHYW1qYSBGbG93ZXIiLCIzIjoiUGVybWFuZW50IE1hcmtlciIsIjQiOiJQYXRyaWNrIEhhbmQiLCI1IjoiSnVzdCBNZSBBZ2FpbiBEb3duIEhlcmUifX0%3D.

How type influences readability – Fonts Knowledge - Google Fonts (06 March 2026). https://fonts.google.com/specimen/Zen+Maru+Gothic/about?query=zen+ma&sort=relevance&lang=en_Latn&script=Latn&preview.script=Latn&preview.lang=en_Latn.